

II. Fractional Calculus and (Quantum) Optics

Fractional Lagrangian Optics — Scientific Expansion (GitHub-Safe Markdown)

1. Introduction

In classical geometric optics, the trajectory of a light ray is governed by **Fermat's principle**, which states that light travels along the path that extremizes the optical path length. This principle can be formulated using **variational mechanics**, where the ray path is obtained by minimizing an action functional.

The action functional is

$$S = \int L d\xi,$$

where ξ is an arc-length parameter and L is the optical Lagrangian.

For a medium with refractive-index matrix $\hat{n}$, the Lagrangian is

$$L = \sqrt{\left(\hat{n} \cdot \frac{d\vec{r}}{d\xi}\right)^2}.$$

The ray position is

$$\vec{r}(\xi) = (r_1(\xi), r_2(\xi), r_3(\xi)).$$

The refractive-index matrix depends on position:

$$\hat{n} = \hat{n}(\vec{r}(\xi)).$$

This introduces nonlinearity into the ray equations.

2. Geometry and the Refractive-Index Matrix

The refractive-index matrix is related to the optical metric tensor:

$$(\hat{n})^2 = \hat{g}.$$

In isotropic media:

$$\hat{n} = nI, \quad \hat{g} = n^2I.$$

The ray derivative is

$$\vec{r}' = \frac{d\vec{r}}{d\xi}.$$

Because $\hat{n}$ depends on $\vec{r}$, the geometry and ray trajectory are coupled.

3. Euler–Lagrange Equations in Optical Form

Extremizing the action yields the Euler–Lagrange equations:

$$\frac{d^\mu}{d\xi^\mu} \left(\frac{\partial L}{\partial \dot{r}_j} \right) - \frac{\partial L}{\partial r_j} = 0, \quad j = 1, 2, 3.$$

For classical optics, $\mu = 1$.

These equations define the ray path for a given geometry $\hat{g}$.

Two interpretations:

- **Forward problem:** Given $\hat{g}$, compute $\vec{r}(\xi)$.
- **Inverse problem:** Given $\vec{r}(\xi)$, deduce $\hat{g}$.

The inverse problem is central in metamaterial design.

4. Fractional Generalization

Fractional calculus generalizes differentiation to non-integer order.

The fractional Euler–Lagrange equations are

$$\frac{d^\mu}{d\xi^\mu} \left(\frac{\partial L}{\partial \dot{r}_j} \right) - \frac{\partial L}{\partial r_j} = 0, \quad \mu \in \mathbb{R}.$$

Fractional derivatives introduce **nonlocality** and **memory effects**.

5. Question 1 — Fractional Derivative Order and Ray Dynamics

5.1 Conceptual Implications

Fractional derivatives imply that the ray evolution depends on its entire past trajectory.

This models media with:

- long-range correlations
- memory
- fractal microstructure

5.2 Mathematical Consequences

Different fractional derivatives exist:

- **Riemann–Liouville:** strong historical dependence
- **Caputo:** convenient for initial-value problems

Fractional Euler–Lagrange equations become integro-differential.

5.3 Physical Interpretation

Fractional optics may describe:

- nonlocal refractive response
- anomalous refraction

- effective media with fractal geometry
- generalized Fermat principles

Ray paths may satisfy fractional extremal conditions rather than classical ones.

6. Question 2 — Mapping Between Geometries via Fractional Order

6.1 Hypothesis

Is there a fractional order μ^* such that a ray in geometry $\hat{g}$ corresponds to a ray in geometry $\hat{g}'$?

6.2 Mathematical Framework

Let $\mathcal{E}_1$ be the classical Euler–Lagrange operator and $\mathcal{E}_\mu$ its fractional version.

We ask whether

$$\mathcal{E}_{\mu^*}[\hat{g}] = \mathcal{E}_1[\hat{g}'].$$

If true, fractional calculus acts as a **geometric transformer**.

6.3 Physical Interpretation

Possible consequences:

- fractional curvature deformation
- optical duality between local and nonlocal media
- metamaterial equivalence
- fractional conformal transformations

6.4 Example Scenario

Metrics:

$$\hat{g} = (\hat{n})^2, \quad \hat{g}' = (\hat{n}')^2.$$

Fractional mapping:

$$E_{\mu^*}[\hat{g}, r] = E_1[\hat{g}', r'].$$

Fractional order modifies effective curvature.

6.4.1 Fractional Curvature Deformation

Fractional operators may emulate:

- stretching/compressing space
- anisotropy
- geodesic curvature changes

6.4.2 Fractional Conformal Transformations

Possible scaling:

$$\hat{g}' = \lambda(\mu^*) \hat{g}.$$

6.4.3 Fractional Effective Media

Fractional order may correspond to:

- sub-wavelength structuring
- fractal microgeometry
- nonlocal electromagnetic response

6.4.4 Ray Equivalence

Ray equivalence condition:

$$r(\xi) \in \mathcal{R}(\hat{g}) \iff r(\xi) \in \mathcal{R}_{\mu^*}(\hat{g}').$$

6.4.5 Implications for Optical Design

Fractional calculus may enable:

- curved-space optics in flat media
- metamaterials mimicking gravitational lensing
- fractional lenses
- tunable optical geometry

7. Question 3 — Consequences for Matrix Optics

Matrix optics uses ABCD matrices:

$$\begin{pmatrix} r' \\ \theta' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r \\ \theta \end{pmatrix}.$$

Fractional dynamics modify this structure.

7.1 Fractional ABCD Matrices

Entries become functions of μ :

$$\begin{pmatrix} A(\mu) & B(\mu) \\ C(\mu) & D(\mu) \end{pmatrix}.$$

7.2 Fractional Paraxial Approximation

Fractional optics may yield:

- fractional Gaussian beams
- nonlocal beam spreading
- modified diffraction
- fractional wavefront curvature

7.3 Fractional Optical Elements

Fractional focal length:

$$f(\mu) = f_0 \lambda(\mu).$$

7.4 Fractional Ray Bundles

Possible effects:

- nonlinear bundle evolution
- memory-dependent divergence
- fractional caustics

7.5 Connection to Fractional Quantum Mechanics

Fractional optics parallels:

- fractional Helmholtz equations
- fractional diffraction integrals
- fractional propagation kernels

8. Conclusion

Fractional calculus introduces nonlocality, memory, and scale-dependent behavior into geometric optics.

Key implications:

- fractional ray curvature
- mapping between optical geometries
- fractional ABCD matrices
- fractional optical elements
- unified framework for metamaterials and fractal optics

This establishes a foundation for future research in fractional optical systems.
